

Nandani Gupta

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EDUCATION

Indian Institute of Technology Jammu, India
Bachelor of Technology in Chemical Engineering

2021 – 2025

WORK EXPERIENCE

Backend Engineer AI/RAG Platform | Lohum Cleantech Pvt. Ltd. (Greater Noida, India)

Tech Stack: Python, FastAPI, LangChain, OpenAI API, Vector Database, PostgreSQL, AsyncIO

July 2025 – Present

- Designed backend services for a **RAG**-based policy intelligence platform indexing **500+** government and regulatory documents into a **vector database** for low-latency **semantic retrieval** and engineered an automated ingestion pipeline in **Python** polling PIB releases hourly (**24×/day**) with document diffing, chunking, and **embedding generation**.
- Built an email ingestion **microservice** monitoring shared mailboxes, parsing communications, and triggering **asynchronous** preprocessing and embedding pipelines using **AsyncIO**, improving indexing throughput and automating policy intelligence surfacing.
- Designed **microservice** components supporting concurrent query processing and high-availability **semantic search** systems.

Software Developer Intern | Arvon Web Solutions (Remote)

Tech Stack: C, C++, Boost.Asio, REST APIs, PostgreSQL, JSON, Docker, Git

May 2024 – July 2024

- Developed backend **RESTful APIs** in **C++** to power a SaaS hospital management platform building modules for patient records, appointment scheduling, and doctor availability, and designing normalized **PostgreSQL** schemas with structured **JSON**-based API responses and referential integrity enforcement.
- Containerized all backend services with **Docker**, managed source control via **Git**, wrote technical documentation for all API modules ensuring reproducible deployments and consistent engineering standards.

INTERNATIONAL EXPOSURE

Research Intern | Hanyang University, South Korea

Tech Stack: Python, MATLAB, NumPy, SciPy, Data Analysis, Scripting

May 2024 – July 2024

- Developed a filter paper-based sensor for detecting chloride ions in concrete to assess corrosion risk, optimizing chemical protocols using **0.01 N Silver Nitrate** and **0.3 N Nitric Acid** replacing toxic reagents with safer alternatives and validating sensor performance via calibration curve modelling and error analysis in **Python** (**NumPy**, **SciPy**) within a cross-functional international research team.

PROJECT MANAGEMENT AND DEVELOPMENT

Fintech Fraud Risk Engine

[GitHub](#)

Tech Stack: Python, Scikit-learn, FastAPI, Pandas, Redis Streams, SQLAlchemy, Docker, SHAP

- Developed a real-time fraud detection platform in **Python** integrating a **Logistic Regression** model (ROC-AUC **0.98**), KYC profiling, behavioral velocity analytics, and **SHAP** explainability, architected as modular **microservices** in **FastAPI** with event-driven ingestion via **Redis Streams** (producer-consumer), **SQLAlchemy**-based persistence, and **Dockerized** deployment.
- Built supervised fraud detection pipelines with **StandardScaler**-based feature preprocessing and probabilistic inference, performed systematic **regression testing** on model outputs.

High-Scale URL Shortener

[GitHub](#)

Tech Stack: Golang, Gin, Redis, PostgreSQL, Bloom Filter

- Built a high-throughput URL shortening service in **Golang** using the **Gin** framework, implementing **Redis**-based caching for low-latency redirection and a **Bloom Filter** for **O(1)** alias existence checks, eliminating cache penetration and sustaining consistent performance under high concurrent traffic.
- Designed a scalable persistence and validation layer using **PostgreSQL** for link metadata storage, achieving efficient lookups through optimized indexing, demonstrating strong **data structures**, **system design**, and algorithmic optimization thinking end to end.

Multiple Elevator System

[GitHub](#)

Tech Stack: C++, OOP, Scheduling Algorithms, System Simulation, Unit Testing, Git

- Designed and implemented a multi-elevator control system in **C++** using **OOP** principles and a score-based scheduling algorithm to efficiently coordinate concurrent floor requests across multiple elevators, validated with a dedicated **unit test** suite covering all state-transition and scheduling edge cases.
- Architected with a modular, header-separated structure (Elevator, ElevatorSystem, Request, Simulator) following clean **software design** principles.

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, SQL, Java

Backend & AI/ML: FastAPI, Node.js, REST APIs, Microservices, LangChain, OpenAI API, RAG, Vector Databases, AsyncIO

Frontend & Infra: React.js, HTML5, CSS3, Docker, Git, PostgreSQL, Redis, SQLAlchemy

Core CS: Data Structures and Algorithms, OOP, System Design, Operating Systems, Computer Networks, Agile/Scrum

PUBLICATIONS & ACHIEVEMENTS

- Solved **450+** data structure and algorithm challenges on **LeetCode**, demonstrating strong logical thinking, problem-solving skills, and persistence across a wide range of difficulty levels.
- Published research in **Springer SCOPUS** on “*Numerical investigation of Spray characteristics of lean-burn gas turbine injector*”, Proceedings of the 10th International and 50th National Conference on Fluid Mechanics and Fluid Power (FMFP), December 2023, IIT Jodhpur. [Publication Link](#)